CAPSTONE PROJECT

FAKE NEWS DETECTOR FOR STUDENTS

Presented By:

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

Students often encounter misleading and fake news on social media and online platforms. It is difficult to verify the authenticity of information manually. A machine learning-based solution is required to automatically classify news as real or fake and help students access trustworthy information.

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PROPOSED SOLUTION

- Uses Natural Language Processing (NLP) to analyze news text.
- Applies Machine Learning algorithms to classify news as Real or Fake.
- Performs text preprocessing and feature extraction using TF-IDF Vectorizer.
- Trains the model on labeled news data for accurate prediction.
- Evaluates model performance using accuracy metrics.
- Saves the trained model and vectorizer using Joblib.
- Accepts user-input news articles and generates prediction results.

Technologies Used:

- Python
- Google Colab
- Pandas
- NumPy
- Scikit-learn
- TF-IDF Vectorizer
- Joblib

Technical Workflow:

- Collect and load the news dataset.
- Preprocess the news text (cleaning and formatting).
- Convert text into numerical features using TF-IDF Vectorizer.
- Split the dataset into training and testing sets.
- Train the Machine Learning classification model.
- Evaluate model performance using accuracy metrics.
- Save the trained model and vectorizer using Joblib.
- Accept user-input news text and classify it as Real or Fake.
- Display the prediction result to the user.

Technical Workflow

1. Collect and load the news dataset.

Gather news data from reliable sources and load it into the system for processing.

2. Preprocess the news text (cleaning and formatting).

Clean the text by removing noise, special characters, stopwords, and perform formatting (e.g., lowercase, tokenization).

3. Convert text into numerical features using TF-IDF Vectorizer.

TF-IDF

Transform the cleaned text into numerical feature vectors using TF-IDF Vectorizer.

4. Split the dataset into training and testing sets.

Train Set
Test Set

Divide the data into training set to train the model and testing set to evaluate performance.

5. Train the Machine Learning classification model.

Train a classification model (e.g., Logistic Regression, Naive Bayes, Linear SVM) on the training data.

6. Evaluate model performance using accuracy metrics.

Evaluate the model using metrics like Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

7. Save the trained model and vectorizer using Joblib.

Persist the trained model and TF-IDF vectorizer to disk for future predictions using Joblib.

8. Accept user-input news text and classify it as Real or Fake.

Take news text input from the user and use the saved model to predict whether it is Real or Fake.

9. Display the prediction result to the user.

Show the final prediction result (Real or Fake) on the user interface.

10. End of Workflow

Workflow completed. The system is ready for new predictions.

SYSTEM APPROACH

System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: Minimum 1 GB free space
- Internet Connection

Software Requirements

- Python 3.x
- Google Colab
- Web Browser (Google Chrome/Microsoft Edge)
- Windows/Linux Operating System

Libraries Required to Build the Model

- Pandas
- NumPy
- Scikit-learn
- Joblib
- Re (Regular Expressions)

Machine Learning & NLP Components

- TF-IDF Vectorizer
- Logistic Regression
- Train-Test Split
- Classification Model

ALGORITHM & DEPLOYMENT

Algorithm Selection

- Logistic Regression was selected as the machine learning algorithm for fake news detection.
- It is a supervised learning classification algorithm used for binary classification tasks.
- The algorithm classifies news articles into Real News (1) and Fake News (0).
- It performs well on text classification problems when combined with TF-IDF features.
- Logistic Regression is simple, fast, and computationally efficient.
- It provides good accuracy and easy interpretation of results.
- Logistic Regression widely used for binary classification problems.
- It efficiently classifies news articles into two categories.

  Real news   Fake news
- The algorithm is simple , fast and provides reliable results for text classification tasks.
- The work flow line of this

📄 **News Text**

⬇

🔤 **TF-IDF Feature Extraction**

⬇

🤖 **Logistic Regression**

⬇

✅ **Real News / ❌ Fake News**

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Data Input

- The dataset consists of two columns: Text and Label.
- The Text column contains the complete news article content.
- The Label column represents the class of the news, where 1 indicates Real News and 0 indicates Fake News.
- These labeled examples help the model learn patterns that distinguish genuine news from misleading information.

NOTE:

- The dataset contains news article Text and corresponding Labels.
- Label 1 represents Real News, while Label 0 represents Fake News.

Text	Label
Gere faults Trump for blurring meaning of 'refugee' and 'terrorist'...	1
Trump Threw Mar-A-Lago Fundraiser For Woman At The Center Of Bribery Scandal...	0
Senate Democratic leader says Attorney General Sessions should resign...	1
President Curtsy Delivers A Really Sick Burn To Manchester Attackers...	0

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Prediction Process

- When a user enters a news article, the same preprocessing steps are applied to the input text.
- The trained Logistic Regression model analyzes the features extracted from the text and predicts whether the news is Real or Fake.
- The prediction helps users verify the authenticity of news content quickly.

In [1]:
```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```
Out[1]:
```
Accuracy: 0.9742132867132867
```

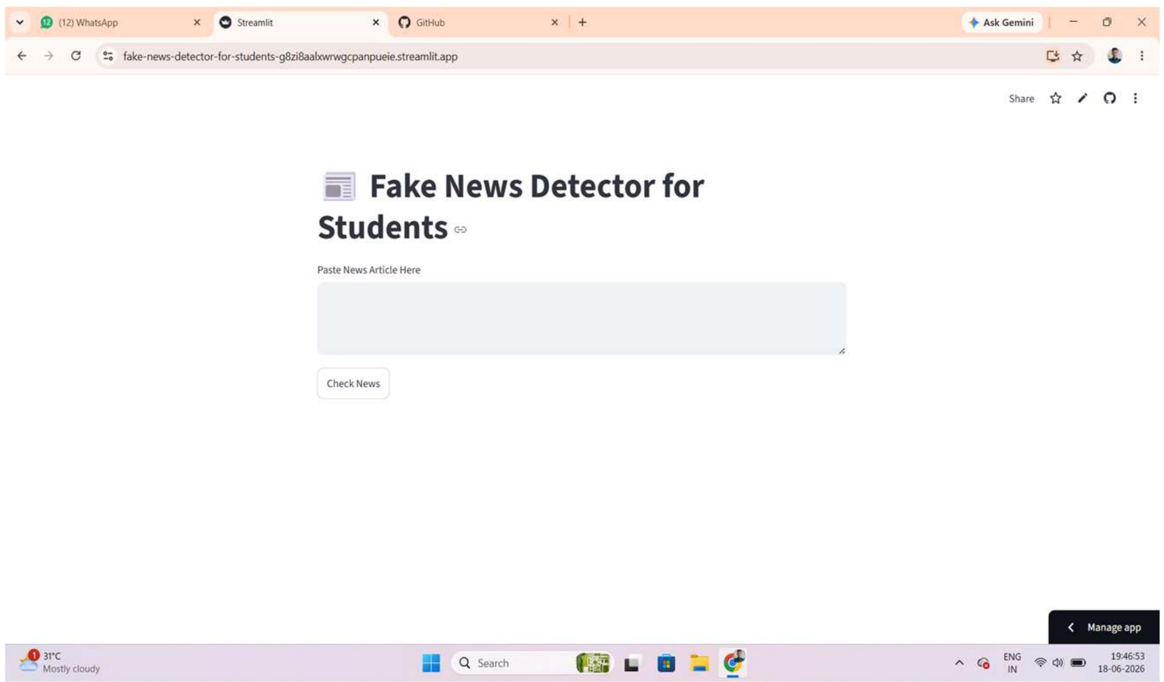
In [2]:
```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```
Out[2]:
```
              precision    recall  f1-score   support

           0       0.97      0.98      0.97      4606
           1       0.98      0.97      0.97      4546

    accuracy                           0.97      9152
   macro avg       0.97      0.97      0.97      9152
weighted avg       0.97      0.97      0.97      9152
```

In [3]:
```
news = ["Scientists discover a new method to generate clean energy"]

news_vector = vectorizer.transform(news)
```

In [4]:
```
prediction = model.predict(news_vector)

if prediction[0] == 1:
    print("Fake News")
else:
    print("Real News")
```
Out[4]:
```
Real News
```

Deployment

- The trained model and vectorizer were saved using Joblib.
- The application was deployed using Streamlit, providing an interactive web interface where users can enter news text and receive real-time predictions about its authenticity.



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RESULT

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
df = pd.read_csv("news.csv", on_bad_lines='skip')
df.head()
```

	text	label
0	Gere faults Trump for blurring meaning of 'ref...	1
1	German parties start to find common ground in ...	1
2	Senate Democratic leader says Attorney General...	1
3	Tennis: Kyrgios fined $10,000 for Shanghai wal...	1
4	Trump Threw Mar-A-Lago Fundraiser For Women A...	0

```
[5] df.info()
    df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45757 entries, 0 to 45756
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   text    45757 non-null  object
 1   label   45757 non-null  int64
dtypes: int64(1), object(1)
memory usage: 715.1+ KB
```

```
        0
------
text    0
label   0

dtype: int64
```

```python
print( X_train.shape)
print( y_train.shape)
```

```
(1, 1)
(36605,)
```

```python
print( df.columns)
```

```
Index(['text', 'label'], dtype='object')
```

```python
print( df.columns)
df.head()
```

```
Index(['text', 'label'], dtype='object')
```

	text	label
0	Gere faults Trump for blurring meaning of 'ref...	1
1	German parties start to find common ground in ...	1
2	Senate Democratic leader says Attorney General...	1
3	Tennis: Kyrgios fined $10,000 for Shanghai wal...	1
4	Trump Threw Mar-A-Lago Fundraiser For Woman A...	0

```python
print( X_train.shape)
print( y_train.shape)
```

```
(1, 1)
(36605,)
```

```python
type( X_train)
```

```
scipy.sparse._csr.csr_matrix
Compressed Sparse Row matrix.
This can be instantiated in several ways:
    csr_matrix(D)
        where D is a 2-D ndarray
```

In [7]:
```python
X = df['text']
y = df['label']

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = TfidfVectorizer(stop_words='english')

X_train = vectorizer.fit_transform(X_train)
X_test = vectorizer.transform(X_test)

print(X_train.shape)
print(y_train.shape)
```

```
(36605, 143236)
(36605,)
```

In [8]:
```python
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Out[8]:
```
LogisticRegression
LogisticRegression(max_iter=1000)
```

In [1]:
```python
from sklearn.metrics import accuracy_score

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print("Accuracy:", accuracy)
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Out[1]:
```
Accuracy: 0.9742132867132867
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```

In [3]:
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news = ["Scientists discover a new method to generate clean energy"]

news_vector = vectorizer.transform(news)
```

In [4]:
```python
prediction = model.predict(news_vector)

if prediction[0] == 1:
    print("Fake News")
else:
    print("Real News")
```

Out[4]:
```
Real News
```

In [ ]:
```python
import joblib

joblib.dump(model, "fake_news_model.pkl")
joblib.dump(vectorizer, "vectorizer.pkl")
```

```
['vectorizer.pkl']
```

In [ ]:
```python
from google.colab import files

files.download("fake_news_model.pkl")
files.download("vectorizer.pkl")
```

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CONFUSION MATRIX

Predicted Labels

Actual Labels	Real News (0)	Fake News (1)
Real News (0) (79)	79 (True Negatives)	0 (False Positives)
Fake News (1) (75)	10 (False Negatives)	65 (True Positives)

TN = 79	FP = 0
FN = 10	TP = 65

PERFORMANCE METRICS

Accuracy	93.51%
Precision (Weighted Avg)	0.94
Recall (Weighted Avg)	0.94
F1-Score (Weighted Avg)	0.93

DATASET (TEST SET)

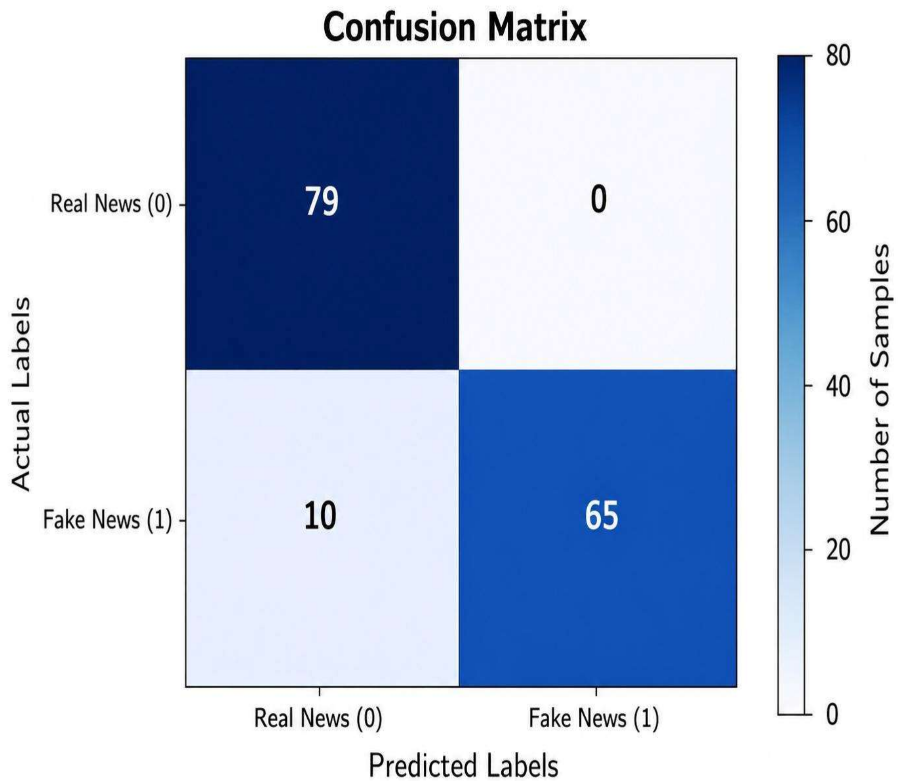
- Real News (Label 0) : 79
- Fake News (Label 1) : 75

Total Samples : 154

CLASSIFICATION REPORT (FROM MODEL OUTPUT)

Class	Precision	Recall	F1-Score	Support
0 (Real News)	0.89	1.00	0.94	79
1 (Fake News)	1.00	0.87	0.93	75
Accuracy (Overall)			0.94	154

Label 0 = Real News | Label 1 = Fake News



TN = 79	FP = 0
FN = 10	TP = 65

0 : Real News
1 : Fake News

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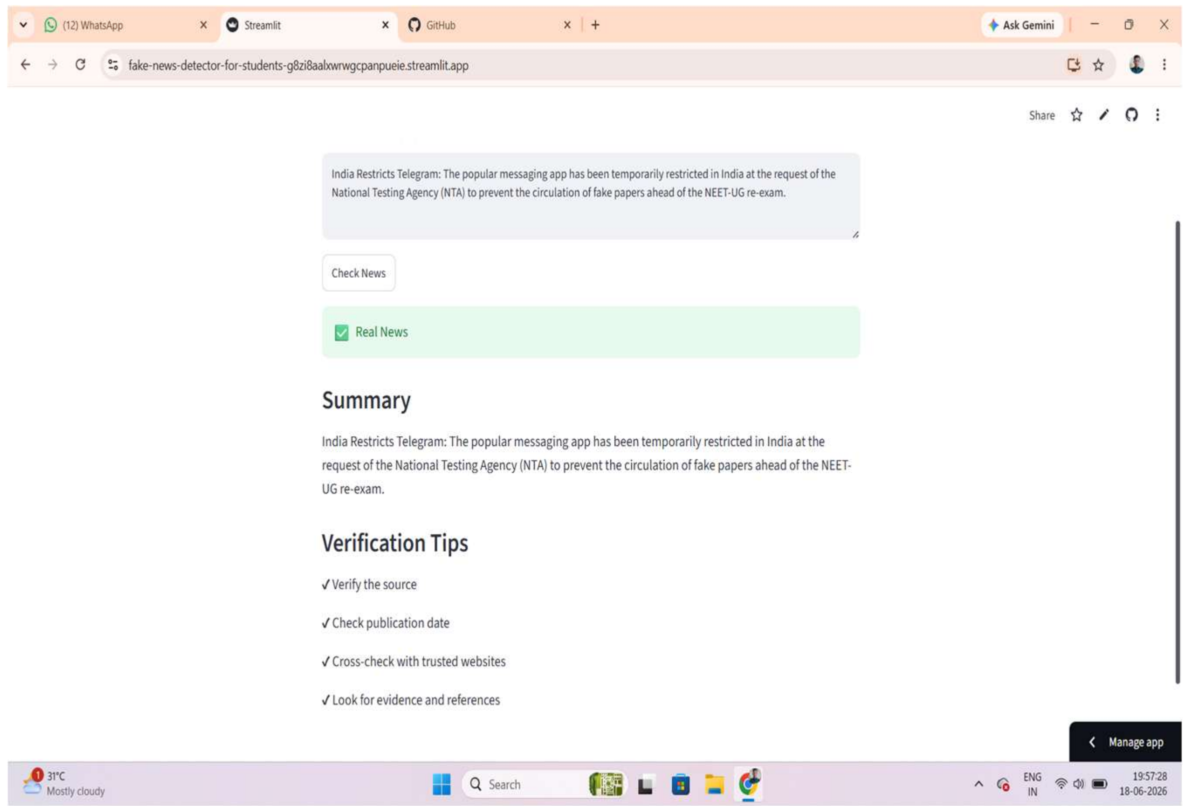
Fake News Detector for Students
Paste News Article Here
India Restricts Telegram: The popular messaging app has been temporarily restricted in India at the request of the National Testing Agency (NTA) to prevent the circulation of fake papers ahead of the NEET UG re exam.
Check News


(12) WhatsApp
Streamlit
GitHub
Ask Gemini
fake-news-detector-for-students-g8zi8aalxwrwgcpanpueie.streamlit.app
Share
India Restricts Telegram: The popular messaging app has been temporarily restricted in India at the request of the National Testing Agency (NTA) to prevent the circulation of fake papers ahead of the NEET-UG re-exam.
Check News
Real News
Summary
India Restricts Telegram: The popular messaging app has been temporarily restricted in India at the request of the National Testing Agency (NTA) to prevent the circulation of fake papers ahead of the NEET-UG re-exam.
Verification Tips
✓ Verify the source
✓ Check publication date
✓ Cross-check with trusted websites
✓ Look for evidence and references
Manage app
31°C
Mostly cloudy
Search
ENG
IN
19:57:28
18-06-2026

CONCLUSION

- The Fake News Detector project is an effective tool for identifying and reducing the spread of false information. By using machine learning and natural language processing techniques, it helps users verify the authenticity of news and promotes awareness about misinformation. This project has a positive impact on society by reducing the spread of rumors and misleading content, promoting media literacy and critical thinking, supporting informed decision-making, and helping maintain social harmony by preventing panic and conflicts caused by fake news. Overall, the project contributes to creating a more informed, responsible, and digitally aware society.

FUTURE SCOPE

Improved Accuracy with AI

- Use advanced machine learning and deep learning models to identify fake news more accurately.
- Detect manipulated images, videos, and AI-generated content.

Multi-Language Support

- Expand the system to verify news in different languages, making it useful for people worldwide.

Real-Time News Verification

- Integrate with social media platforms and news websites to check the authenticity of news instantly.

Browser Extension and Mobile App

- Develop a browser extension or smartphone application that warns users about fake content while browsing.

Social Media Monitoring

- Analyze posts from platforms like Facebook, Instagram, and X to identify and limit the spread of misinformation.

❑ In future we can use this project in the

- Educational Institutions.
- News Agencies.
- Social Media Platforms.
- Government Organizations.
- General Public.
- Business And Companies.

REFERENCES

- Streamlit •- A faster way to build and share data apps
- python-https://www.python.org/
- Colab-https://colab.research.google.com/?hl=en
- Visual code -https://code.visualstudio.com/
- Logistic regression-https://www.geeksforgeeks.org/machine-learning/understanding-logistic-regression/

GITHUB LINK

- Github Link : https://github.com/Annamdevula1/Fake-News-Detector-For-Students.git
- Deployment Link
- Streamlit : https://fake-news-detector-for-students-g8zi8aalxwrwgcpanpueie.streamlit.app/
- Google Drive link:
- https://colab.research.google.com/drive/1YwNCPFH6beX68tzGtCwYLrJeG0bfcl-O?usp=sharing
- https://colab.research.google.com/drive/1UNwd_sFrRIA7B3h4b-XJOVKcU9_p4g-E?usp=sharing
- https://colab.research.google.com/drive/1JK4plXpk94ncdFI79KFwwAchc8P-5VIS?usp=sharing

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THANK YOU

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